Singapore

Tonight the Central region's 24-hour PSI is 108. The El Niño peak is Singapore's safest stretch; the danger is haze now and heat after it. The deaths this event can be counted as causing come from haze and dengue, and the losses nobody insures are workers' hours.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Singapore, national (spans roughly 20 Climate Impact Lab regions)	No figure published for Singapore	Jun 2026 – Feb 2027 (heat only)	Transboundary haze now; dengue; heat Feb–May 2027

18/18 recommendations verified. 8 corrected (partially confirmed), 1 confirmed, 3 unverified, 2 redundant, 4 contraindicated. The funding target moved from SGD 600k to SGD 425k, a 29% cut, under a third, so the envelope stands; the ceiling fell from SGD 1.2M to SGD 700k. **Resolution caveat:** Singapore was given as a whole country. The method is built for sub-city places, so this briefing names the estates, housing types, and groups where actions differ, but its numbers are national. **Method caveat:** the session's web-search budget was exhausted; every agent worked by direct fetch. Unreachable this run: MOH pages (so the haze medical subsidy is unconfirmed), SingStat and HDB counts, MSS annual climate statements, and several news sites. Exposure figure: Climate Impact Lab, Sep 2026, pre-peer-review. Its Southeast Asia figures name the Philippines, Vietnam, Thailand, and Cambodia together (19,400) and Indonesia (19,300). Nothing here is scaled from them.

EXECUTIVE SUMMARY — AT A GLANCE



1.29°N 103.85°E, Singapore · Natural Earth

- **The risk.** Haze comes first and heat follows in February to May 2027: in 2013 the 3-hour PSI reached 401, and the 2005 dengue epidemic brought 14,209 cases and 27 deaths.
- **First action.** From PSI 51 through November, AIC, the Silver Generation Office, and Active Ageing Centres reach every person over 65 living alone in a rental block with a mask fitting, a sealed room, and a named neighbour.
- **The money.** SGD 250k (USD 194k) floor, SGD 425k (USD 329k) target, SGD 700k (USD 543k) ceiling; private money saves lives only where state schemes stop, at the work-permit line and inside one-room rental flats.
- **Do not.** Do not make mass N95 handouts the haze plan.

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* Suggested formulation: check with a health worker before sharing.

EXPOSURE, HONESTLY

WHAT IS AND IS NOT QUANTIFIED

- **Haze, live tonight.** At 21:00 on 26 September the 24-hour PSI read 108 in the Central region, in the unhealthy band, and 76 to 93 elsewhere; PM2.5 ran 35 to 62 μg/m³ (data.gov.sg). The ASEAN Specialised Meteorological Centre has the southern ASEAN region at alert Level 3, with moderate to dense smoke over Kalimantan and Sarawak. The Southwest Monsoon runs into early October and haze can persist through the October–November inter-monsoon. **The window is now to about November.**
- **What haze does to the heart here.** Singapore studies: out-of-hospital cardiac arrest RR 1.37 on unhealthy PSI days and 1.10 on moderate days, strongest over 65 (n=8,589); stroke IRR 1.14 at PSI 101 and above and 1.10 at 51 to 100 (n=29,384); heart attack IRR 1.04 per 30 PSI units (n=53,948). **The risk starts at moderate PSI, from 51**, not at the unhealthy line.
- **Heat is not measured to kill here.** No Singapore curve links temperature to all-cause deaths. The best local cardiac-arrest study (Ho 2025, 27,209 cases) found a higher cumulative risk at 24 °C (RR 1.33, 1.04–1.70) than at 29 °C (RR 1.13), and a stronger effect from humidity. Heat does show up in emergency-department admissions and in outdoor work. The 37.0 °C record fell on 17 April 1983 and was tied on 13 May 2023; the 2023 record came *before* that El Niño's onset, which weakens the claim that heat reliably lags the peak.
- **Dengue.** 68 cases in the partial week to 25 September; 2,411 in 2026 to week 37; 11 active clusters, the largest a 74-case red-alert cluster at Ho Ching Road and Kang Ching Road. Risk peaks near a 32 °C maximum and rises once Niño 3.4 passes about +1.4, at a four-week lag. The 2020 record year (35,315 cases, 32 deaths) followed the re-emergence of DENV-3 after thirty years.
- **Who is exposed.** 21.4% of citizens are 65 or over and 152,000 are 80 or over (Population in Brief 2026). 490,700 Work Permit holders work in construction, shipyards, and process industries, and 323,800 are migrant domestic workers (MOM, June 2026). Counts of seniors living alone in one- and two-room rental flats could not be fetched.

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected harm averted per unit of effort, with haze actions first because the hazard is live tonight. Four lenses independently said

* Suggested formulation: check with a health worker before sharing.

the November–January peak is the wrong deadline: the Northeast Monsoon washes out haze in those months.

From PSI 51, reach every person over 65 living alone in a rental block: an assisted mask fitting, a sealed room, and a named neighbour, delivered through Active Ageing Centres and Silver Generation Ambassadors

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · FRONTLINE · SKEPTIC · PHILANTHROPIST

ADAPTATION INVENTORY: NO HAZE ENTRY; CLOSEST, US HEAT ADVISORY THRESHOLD

OWNER	BY WHEN	COST	PREVENTS
AIC, Silver Generation Office, Active Ageing Centre operators, CDCs	Now, through November	Staff and volunteer time; kits see funding	Cardiac arrest and stroke on haze days (RR 1.10 at moderate, 1.37 at unhealthy)

Two corrections shaped this card. The cardiac evidence covers everyone over 65, not only people with known heart disease, and it starts at moderate PSI. And in a Singapore household sample only 12.6% of people wore an N95 correctly, with picture instructions making no difference, so a mask handed over at a door protects about one person in eight. The fitting has to be done in person. HEPA purifiers cut indoor smoke 55 to 65% in controlled studies but only 15% in homes during the 2025 Los Angeles fires, and no trial shows fewer heart attacks or deaths; a purifier-cleaned room is a reasonable measure, not a proven one.

Run it through the existing channels. Active Ageing Centres and Silver Generation Ambassadors already visit, and a separate volunteer door-knock risks duplicating them. The Frontline lens's warning belongs in the brief to every visitor: the senior most at risk does not answer an unknown number, speaks Hokkien, Teochew, Cantonese, or Malay, and switches the fan off to save on the electricity bill. In 2015 only 29,000 elderly received AIR+ masks and care packs.

Make MOM's haze guidance binding: stop heavy outdoor work above PSI 300 and defer it from 201

PARTIALLY CONFIRMED | EMERGENCY MGR · ECONOMIST | ADAPTATION INVENTORY: PRECEDENT, US WORKPLACE HEAT MANDATE

OWNER	BY WHEN	COST	PREVENTS
MOM with NEA	15 October 2026	Staff time	A 2013-style judgement on the day while crews breathe it

* Suggested formulation: check with a health worker before sharing.

MOM already publishes haze guidance by PSI band: at or below 100, 101 to 200, 201 to 300, above 300, and above 400. It says to minimise and defer outdoor work above 300 and to use full-face respirators above 400, but **none of it is mandatory**. The action is narrower than the lenses proposed: turn the existing bands into a requirement. School closures are already set at PSI 300 with exams moved indoors, so no new school protocol is needed.

Employers adopt a written policy that heat-rest minutes are paid, before the rules take effect on 1 December

CONFIRMED GAP | ECONOMIST · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: PRECEDENT, US WORKPLACE HEAT MANDATE

OWNER	BY WHEN	COST	PREVENTS
Main contractors, shipyards, process plants; MOM to follow	1 December 2026	Wage cost to employers; no public money	Workers skipping required rest to protect their pay

Confirmed on MOM's own documents. Rest under shade is already required for heavy physical work: at least 10 minutes an hour at WBGT 32 to under 33, and at least 15 minutes an hour at 33 and above. From 1 December 2026 training, cool water, emergency cooling, and protective clothing become mandatory too. Cooled or well-ventilated rest areas remain only recommended, and **nothing in either document says the rest is paid**. A rest a worker cannot afford to take is a rest on paper. The Adaptation Inventory's US workplace mandate is the closest precedent: heat injuries on very hot days fell sharply after it, without major labour-market disruption.

Worksite WBGT meters are already required on construction sites with contracts of S$5M or more, shipyards, and process plants, with hourly monitoring, and NEA runs about 30 public stations. The Skeptic's proposal of sensors is therefore redundant; its narrower point, independent readings that trigger unannounced inspections, could not be checked against MOM practice.

Delivery platforms cover the minutes riders wait outdoors, on the High heat band or 24-hour PSI above 200

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Grab, foodpanda, Deliveroo and their insurers	1 December 2026	~$20/day payment proposed (lens estimate, unpriced)	Heat and haze exposure in the uncovered gap

* Suggested formulation: check with a health worker before sharing.

The gap is exact and confirmed. Work injury compensation has covered platform workers since 1 January 2025, with medical claims up to $53,000, from travel to pick-up until drop-off. It excludes injuries "while waiting for tasks," which is when riders sit outdoors. Whether any operator already pays heat or haze support could not be checked. This needs the operators to agree, so it is a request, not a rule.

Put Wolbachia into the current red-alert cluster and the towns not yet covered, and watch for a serotype switch

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
NEA	Before the early-2027 heat	Within NEA's programme	A 2020-scale year: 35,315 cases, 32 deaths

The best evidence is now a cluster-randomised trial (Lim et al., NEJM 2026): 71 to 72% efficacy at 3 to 12+ months of exposure. Earlier target-trial emulations gave 47% at three months and 61% at twelve. **Coverage is the binding limit.** The two figures the lenses quoted measure different things: about 1 million residents, roughly 20%, in 2023, and a target of 800,000 households, about 50%, by the end of 2026. Current coverage sits somewhere between. Ang Mo Kio starts in October; Toa Payoh started in July. Whether the Ho Ching Road cluster lies in a covered area, and whether production can speed up in months, could not be checked. A serotype switch preceded both the 2020 record and the 2022 surge, so surveillance is the cheap early signal.

The Skeptic's paradox stands: Singapore has held the Aedes house index near 1% since the 1990s and still had its worst year in 2020, because low population immunity lets a new serotype run. More household checks meet diminishing returns.

Heat and haze voice notes in Bengali, Tamil, and Burmese through MWC's peer ambassadors, and in Tagalog and Bahasa Indonesia for domestic workers, before 1 December

PARTIALLY CONFIRMED | FRONTLINE · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Migrant Workers' Centre ambassadors, dormitory operators, ItsRainingRaincoats	1 December 2026	Volunteer time	Workers not knowing heat-stroke signs or their right to rest

* Suggested formulation: check with a health worker before sharing.

Feasible and not redundant: MWC runs a 24-hour helpline, a chatbot, and a peer ambassador network, and carries no heat content today. Voice travels further than text. One correction: the Sunday hubs at Lucky Plaza and City Plaza are for Filipino and Indonesian domestic workers, so the South Asian languages miss them. The dormitory figure the lens used, about 200,000 workers in 43 purpose-built dormitories, is from around 2020 and probably low.

Give the heat money a February–May 2027 deadline, not a November one

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Every funder and agency planning heat support	Funds live by 1 February 2027	Timing only	Money arriving in the wet, safe months and gone by April

Both 37.0 °C records fell in the April–May inter-monsoon and the Northeast Monsoon runs December to early March. The lag argument is weaker than the lenses said: the 2023 record came before that El Niño began, and the annual climate statements that would test 1998 and 2016 could not be fetched. The seasonal logic holds without it.

A pre-approved U-Save top-up paid automatically on a heat or haze trigger to one- to three-room HDB and Silver Support households

UNVERIFIED | ECONOMIST | ADAPTATION INVENTORY: PRECEDENT, MEXICO INCOME SUPPORT REDUCED HOT-DAY MORTALITY

OWNER	BY WHEN	COST	TRIGGER
MOF	Needs Cabinet approval; slow	Not sized	10+ High heat days in a month, or 24-hour PSI above 100 for 3 days

* Suggested formulation: check with a health worker before sharing.

The channel exists: U-Save is paid quarterly without application, at 1.5 times the usual rate up to $570 this year, with $500 CDC Vouchers per household in January 2027. Both are fixed amounts on a fixed calendar. No weather-triggered utility support has any precedent in Singapore. Mexico's income and tax reform reduced mortality on days above 30 °C by up to 2.75%, largely through electricity spending, which is the mechanism here; the setting differs.

Switch on MOH's haze medical subsidy at a published PSI line, not after the haze arrives

UNVERIFIED | EMERGENCY MGR · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
MOH	Now	Existing scheme	A 1997-style surge (+30% attendance) before the subsidy switches on

Every MOH page returned an error this run. The scheme's terms as the lenses recalled them (about $10 at CHAS clinics for seniors, children, and lower-income patients, activated in 2013, 2015, and 2019 at the Ministry's discretion) are unconfirmed. Check moh.gov.sg before citing any of it.

Make cooled or ventilated rest areas a permit condition for large sites

UNVERIFIED | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
BCA with MOM	Not before this season	Contractors; net cost to the state likely negative	Rest taken in heat that does not cool

Cooled rest areas are recommended, not required, and no sign was found that BCA plans to change that. Realistic in principle, unlikely before the peak.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	AIC & Silver Generation Office	Active Ageing Centres	CDCs	MOM	NEA	Contractors, shipyards & plants	Delivery platforms & insurers	MWC & ItsRainingRaincoats	Dormitory operators	MOF	MOH	BCA	Funders (incl. Tote Board)
1. Reach seniors living alone from PSI 51	**A/R**	**R**	**R**										**C**
2. Make MOM's haze guidance binding				**A/R**	**C**	**I**							
3. Paid heat-rest policy before 1 December				**I**		**A/R**							
4. Platforms cover riders' outdoor waiting time							**A/R**						
5. Extend Wolbachia; watch for serotype switch					**A/R**								
6. Heat and haze voice notes, five languages								**A/R**	**R**				
7. Heat money live February to May 2027													**A/R**
8. Automatic U-Save top-up on a trigger					**C**					**A/R**			
9. Haze medical subsidy at a published PSI					**C**						**A/R**		
10. Cooled rest areas as a permit condition				**C**		**I**						**A/R**	

What the chart shows. **MOM is accountable for only one of the three worker-protection rows.** It owns the binding haze guidance, but on paid rest it is only "to follow", and cooled rest areas sit with BCA, so heat-rest protection depends on employers moving first. **Rows 3 and 4 have no public owner.** Paid rest and riders' waiting time rely on contractors and delivery platforms agreeing, so they are requests, not rules. **Rows 1 and 6 put volunteers at the door.** Silver Generation Ambassadors and MWC's peer ambassadors do the reaching, on staff and volunteer time. **Row 7 has no single owner in the report.** It is assigned to every funder and agency planning heat support, so the A is placed on funders, who control when the money goes live.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled and verified. In a state this well funded, private money saves lives only where state schemes stop: at the work-permit line, and behind the doors of one-room rental flats.

FLOOR

TARGET

CEILING

LARGEST UNCERTAINTY

* Suggested formulation: check with a health worker before sharing.

SGD 250k (USD 194k)	SGD 425k (USD 329k); 500k with match	SGD 700k (USD 543k)	Unit cost of a clean-room kit; Tote Board match conditions

Arithmetic at 1 SGD = 0.775 USD. Floor: TWC2 150k + Red Cross reserve 100k = SGD 250k. Target: 150k + Lions Befrienders 100k + HealthServe 75k + Red Cross 100k = SGD 425k, or 500k with the Tote Board's 75k match. Ceiling: TWC2 250k (reaching the 125k match cap) + Lions Befrienders 150k + HealthServe 100k + Red Cross 200k = SGD 700k. The target fell 29% from the lens's SGD 600k, under a third, so the envelope stands. The lens's SGD 1.2M ceiling was not supportable: TWC2's whole appeal is SGD 500k over six months, HealthServe runs one clinic, and the seniors' placement had no costed basis. Dengue gets SGD 0 from this envelope because Wolbachia is state-run; see the disagreement below.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
TWC2 (CareFund, Cuff Road meals)	Migrant workers sick or injured and off pay, including heat illness	SGD 150k	Donation matched 50% by the Tote Board, cap SGD 125k, to 31 Mar 2027	Appeal target SGD 500k; only SGD 8,955 raised by 20 Sep, so the match has room. 2025 spend on the two programmes about SGD 274.5k, so 150k is about 55% of a year. General relief, not heat-specific. Whether a single large donor qualifies for the match is unconfirmed.	Keep
Singapore Red Cross	Contingent reserve for haze and heat relief	SGD 100k	Released at PSI above 200 or an NEA High heat advisory	Live delivery channel: a heat-and-haze care-pack appeal ran April to July 2026 for 10,000 people, including seniors alone and outdoor workers. The fastest route to help by 31 October. Donations carried a 250% tax deduction.	Keep
Lions Befrienders (10 Active Ageing Centres)	Assisted mask fitting and purifier clean-room kits for seniors over 65 living alone	SGD 100k, only against a quote	Restricted grant	Redesigned from fans, which the Red Cross packs already cover. No purifier, fan, or per-visit cost could be fetched; the lens's 2,000 × SGD 100 was an assumption. Purifier benefit is shown in indoor PM2.5 and biomarkers, not in deaths.	Cut, hold

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
HealthServe	Heat-illness outreach and clinic hours for dormitories, Feb–May 2027	SGD 75k	Direct grant	One clinic, three evenings a week; about 5,300 consultations and 14,200 workers reached in 2024. Budget unpublished, so absorption is unknown. No heat programme on record.	Cut, hold until annual report read
Dengue outreach	—	SGD 0	—	State-funded Wolbachia; household checks meet diminishing returns at a 1% house index.	Agreed, contested

The funder's one thing survives verification in a narrower form: the state out-spends any private fund, so the grant dollar belongs where the state's schemes stop. Here that means wages lost to heat and illness for work-permit holders, and a trained person at the door of a senior who will not answer the phone. Honest expectation: deaths averted by this envelope are in single digits, so the cost per death averted is likely above USD 100k. That is still the right use of money that has nowhere better to go in this city.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for heat and haze and would fail or harm here.

Do not make mass N95 handouts the haze plan

CONTRAINDICATED SKEPTIC; CONFIRMED

In 2013 masks sold out at PSI 401 and MOH drew on nearly 9 million from its stockpile, distributing more than 4 million. Supply was fixed. Fit never was: in a 2015 Jurong household study only 12.6% passed a visual fit check, and instructions did not help. A handout buys false reassurance for most of the people who receive it.

* Suggested formulation: check with a health worker before sharing.

Instead: assisted fitting for the over-65s, and one sealed, purifier-cleaned room per high-risk household.

Do not tell older people to avoid fans*

`CONTRAINDICATED` `SKEPTIC; CONFIRMED WITH CORRECTIONS`

The "fans are dangerous" warning comes from hot, dry climates. Fans became harmful around 39 °C in a study of 20 young adults, and only when they were dehydrated; dehydration had two to two and a half times the effect. In older adults at 38 °C and 60% humidity, a fan lowered core temperature slightly; harm appeared at 45 °C and 15% humidity. Singapore's typical maximum is 31 to 32 °C and its record is 37.0 °C, in humid air.

Instead: "Use a fan, drink water, and move to air-conditioning above about 38 °C or if unwell."

Do not build cooling centres and wait for isolated seniors to come

`CONTRAINDICATED AS THE DEFAULT` `SKEPTIC; CONFIRMED ABROAD, UNTESTED HERE`

In Maricopa County, 49% of the public did not know cooling centres existed and 31% of visitors cited transport as a barrier. In Virginia fewer than 7% of people below the poverty line lived within a 15-minute drive of one, against more than 65% overall. Heat deaths happen at home, often at night, when centres are shut. No Singapore pilot was found.

Instead: the door, through the Active Ageing Centre, in the senior's own language.

Do not frame Singapore's El Niño risk as heat deaths at the November–January peak

`CONTRAINDICATED` `PUBLIC HEALTH · EMERGENCY MGR · HISTORIAN`

There is no Singapore heat–mortality curve and no Climate Impact Lab figure for Singapore. The peak months are the wettest and haze-free. A plan built on "heat deaths, November to January" misses the haze that is here tonight and arrives too early for the heat in April.

* Suggested formulation: check with a health worker before sharing.

DROPPED AS REDUNDANT

ALREADY IN PLACE

- **Run NEWater and desalination at full output from late 2026.** That is PUB's own operating lever, not a community action. NEWater meets about 30% of demand and desalination up to 30%; the long-term target is 50% NEWater by 2060, not 40% by 2030 as a lens said. The Linggiu Reservoir low of 2016 and February 2014 as the driest month could not be sourced.
- **Set PSI triggers for school closures and indoor exams.** Already in place: schools close at PSI 300 and above, and national exams move indoors with purifiers.

WHAT HAS HAPPENED HERE BEFORE

THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
17 Apr 1983	37.0 °C, national record, in the 1982–83 El Niño year	—	—
1997–98	El Niño haze	Haze-related hospital attendance +30%; US$0.4bn damage with Malaysia	ASEAN Agreement on Transboundary Haze Pollution, 2002
2005	Dengue epidemic	14,209 cases, 27 deaths	S$7.5m on top of an existing S$2.5m budget; 4,200 volunteers
21 Jun 2013	3-hour PSI 401, first "hazardous" reading	Masks sold out	MOH released over 4M N95s from a ~9M stockpile
25 Sep 2014	Transboundary Haze Pollution Act commences	—	Six firms noticed; no conviction by 2021
Sep 2015	El Niño haze; PSI 341 on 24 Sep; schools closed 25 Sep	No Singapore mortality figure found	AIR+ masks and care packs to 29,000 elderly
Sep 2019	Haze, PSI 154 on 19 Sep	—	16M N95s stockpiled; exams moved indoors

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2020	Dengue record after DENV-3 re-emerged	35,315 cases, 32 deaths	Wolbachia expansion
13 May 2023	37.0 °C record tied, before that El Niño began	—	Heat Stress Advisory and MOM measures
Apr–Jul 2026	Singapore Red Cross heat-and-haze care packs for 10,000 people	—	Fans for seniors partly covered
28 Aug 2026	MOM heat framework enhancements announced	—	Four measures mandatory from 1 Dec 2026; rest unpaid
26 Sep 2026	ASEAN haze alert Level 3; Central PSI 108; dengue cluster of 74 at Ho Ching Road	—	This briefing

The recurrence nobody has noticed, from the Historian and confirmed in part: Singapore's hazards do not arrive at the El Niño peak. Haze comes before it, heat comes after it, and every plan built for "November to January" is early for one and late for the other. And the law written after 2013 to punish the source has convicted no one.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Heat as a killer here** versus **no local evidence it is**. Three lenses built actions around heat; the Public Health lens found no Singapore heat–mortality curve and a cardiac-arrest study where cooler days carried more risk. Resolved toward the evidence: heat actions stay for workers, where admissions and injuries are measured, and drop down the ranking for the general population.
- **Philanthropy should put nothing into dengue** versus **Wolbachia coverage is the binding limit**. Both are true. Coverage will be at most half of households by the end of 2026, which leaves a real gap, but it is a gap only NEA can fill. What would change the call: evidence that a community partner can speed releases, which nobody found.
- **Assisted fitting** versus **purifier clean rooms**. The mask-fit failure is measured here; the purifier's benefit is measured in indoor air and blood pressure, not in deaths. Neither alone is proven; together they are the best available.

* Suggested formulation: check with a health worker before sharing.

- **Volunteer door-knocks** versus **existing programmes**. The Frontline lens wants dialect-speaking volunteers on every floor; verification suggests Silver Generation Ambassadors and Active Ageing Centre befriending already cover the ground. Their scope and heat-season activity could not be fetched, so run volunteers through them, not beside them.

WHAT NOBODY COVERED — BLIND SPOT

No lens looked at domestic workers as a heat-exposed group in their own right: 323,800 people who cannot leave the flat, often caring for the elderly person the rest of this briefing is trying to protect. No lens asked what happens to the over-65s in private flats and landed homes, who sit outside every rental-block programme. No lens considered indoor air in the dormitories themselves on haze days, where ten to twenty men share a room. And no lens could fetch the counts that size the whole effort: how many seniors live alone in one- and two-room rental flats, and how many have no working air-conditioner.

HOW TO READ THIS — METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

* Suggested formulation: check with a health worker before sharing.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

This run. Singapore was given as a whole country, coarser than the method prefers, so the actions name estates, housing types, and groups while the numbers stay national. The session's web-search budget was exhausted, so all thirteen agents worked by direct fetch; the unverified count reflects pages that were blocked, not claims that were refuted. Six verifiers checked the eighteen recommendations; the verdict tags carry the result.

The El Niño projection covers *heat only* and publishes nothing for Singapore. The Adaptation Inventory was consulted through its data feed (96 entries, 30 countries): it holds no Singapore entry and nothing on haze, dengue, or Wolbachia. Its heat entries from elsewhere are cited where they apply; the entry pages redirected for one verifier, so their text was read from the feed.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Singapore. Climate Impact Lab impact region, as the report records it: Singapore, national (spans roughly 20 Climate Impact Lab regions). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.

* Suggested formulation: check with a health worker before sharing.

- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (fans and cooling for older people) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (verification status in brackets) · Live PSI data.gov.sg · ASEAN haze alert asmc.asean.org · monsoon timing weather.gov.sg [confirmed, 26 Sep 2026] · Cardiac arrest and PSI, Ho 2018 europepmc.org · stroke and PSI europepmc.org · heart attack and PSI PMC6475051 · hot and hazy stroke PMC10049712 [confirmed, population corrected] · N95 fit, Singapore households PMC7240349 [confirmed] · purifiers in smoke Barn 2008 · Eaton fire homes [biomarker evidence only] · MOM haze guidelines mom.gov.sg [confirmed, advisory] · MOM heat framework, 28 Aug 2026 mom.gov.sg · measures infographic file.go.gov.sg · heat stress bands weather.gov.sg [confirmed] · Platform worker injury cover mom.gov.sg · Work Permit medical insurance mom.gov.sg · foreign workforce numbers mom.gov.sg [confirmed] · Budget 2026, U-Save and CDC Vouchers singaporebudget.gov.sg · SEADRIF seadrif.org [confirmed] · Sudden cardiac arrest and temperature, Ho 2025 europepmc.org · military heat injury PMC9875869 · work at height in heat PMC12129388 · fans and dehydration PMC12351413 · fans in older adults PMC12308431 [corrected] · Cooling centres, Maricopa PMC12021073 · Virginia PMID 35792811 [confirmed] · Dengue cases nea.gov.sg · Wolbachia nea.gov.sg · coverage 2023 and house index PMC10286981 · target-trial emulation

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PMC11500660 · climate thresholds PMC12727700 · 2020 outbreak wikipedia [confirmed with corrections] · Population in Brief 2026 population.gov.sg · Haze history 1997 · 2013 · 2015 · 2019 · Haze Act wikipedia · climate records wikipedia · 2005 dengue wikipedia · water supply wikipedia [Wikipedia-level; primary sources not opened] · TWC2 appeal twc2.org.sg · Lions Befrienders lionsbefrienders.org.sg · HealthServe healthserve.org.sg · Red Cross heat and haze packs redcross.sg · Active Ageing Centres aic.sg · MWC mwc.org.sg [confirmed] · Unreachable or unconfirmed this run: MOH haze subsidy terms; SingStat and HDB rental and living-alone counts; MSS annual climate statements; Silver Generation Office scope; purifier and fan prices; platform operators' heat support; Linggiu 2016 low; February 2014 driest month.

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions; US workplace heat mandate entry; US heat advisories entry; Mexico income support entry)

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